

An Example of a Metal Printed Antenna

In 2019, Boeing became the first company to print a metal 3D printed antenna for a satellite that was sent to space on AMOS 17 launch vehicle.



Image of metal 3D printed antenna used in AMOS 17 (Credit: Boeing)

“Replacing a design that involved multiple parts bolted together with a single 3D printed unit means fewer parts to assemble and fewer interfaces, which reduces the possibility of issues during manufacturing and testing. Also, the conventional, heavier antenna requires two or three months to manufacture, while the lighter 3D version can be printed in under a week. All that adds up to savings in mass, time and ultimately, costs.”

— **Brian Baker, AMOS 17 Program Manager**

present in other metal 3D printing techniques. Binder jetting does not melt the metal powder, thus eliminating issues related to the build-up of residual stresses. This method does not require a support structure since the loose unused powder gives support to the design.

Advantages of metal 3D printing

Just as all manufacturing processes have their own pros and cons, so does the metal 3D printing technique. Some of the advantages of metal 3D printing over existing manufacturing techniques are:

Cost-effective. 3D metal printing can be used to create intricate parts without requiring any additional moulds, tools, etc. Furthermore, this technique has lower wastage compared to subtractive manufacturing techniques where a larger metal block is shaved off to create the desired metal part.

Customisable. Unlike bulk manufacturing processes, such as forging, which require a mould, it

is resource-intensive for making the custom parts. Metal 3D printing can easily be used to make custom changes to a prototype.

Speeds up design. The traditional manufacturing technique requires expensive tools and machines; hence, most companies need to outsource manufacture of their components—making it both expensive

and time consuming. On the other hand, the 3D printing machines speed up the design process, which can be done in-house.

Eco-friendly. Being an additive manufacturing process, metal 3D printing requires lower energy. Moreover, the material wastage from this process is substantially lower than from any subtractive manufacturing process.

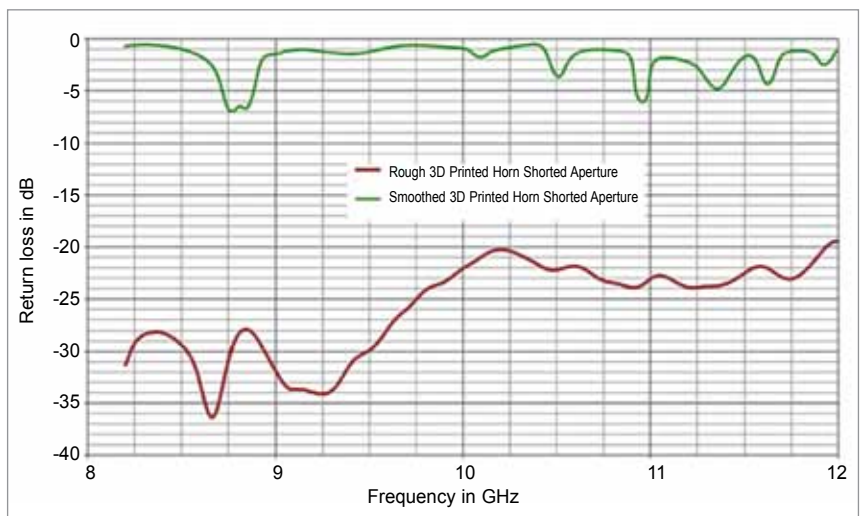
Disadvantages of metal 3D printing

There are some disadvantages as well as, though. Some of these are:

Expensive. Metal 3D printing is extremely expensive for manufacturing objects when large quantities of simple objects need to be manufactured.

Slow. When manufacturing simple and identical designs, other processes, such as forging or many subtractive manufacturing techniques, can create a large number of components in a considerably lower time, since 3D printing makes an object layer by layer, thus consuming more time. Furthermore, post-processing of components is needed to make the final component market ready.

Prone to distortion and warping. Metal 3D printing is prone to



Return loss comparison between rough antenna and smooth antenna after post-processing (Credit: antennatestlab)